

Adapted Equipartition Calculation for the Supernova Remnant J0450–7050

Project III

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Abstract

In this project, an adapted equipartition calculation was applied to estimate the amplified and ambient magnetic fields of the Large Magellanic Cloud supernova remnant J0450–7050. Using the observed radio flux density, spectral index, angular size, distance, and shock velocity, the synchrotron emissivity relation was used to determine the parameter J . Electron and proton equipartition assumptions were then considered separately. The required energy integrals were evaluated numerically and the resulting nonlinear equations were solved using Python. The electron equipartition calculation yielded magnetic fields of $24.12\ \mu\text{G}$ and $7.35\ \mu\text{G}$ for the amplified and ambient fields, respectively, while the proton equipartition calculation gave $77.38\ \mu\text{G}$ and $23.57\ \mu\text{G}$. The results demonstrate the application of adapted equipartition methods to supernova remnants with particle energy indices smaller than two.

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1 Introduction

Magnetic fields are fundamental components of supernova remnants (SNRs) and play an important role in the acceleration and propagation of cosmic rays. Radio observations of SNRs reveal non-thermal synchrotron emission produced by relativistic electrons spiraling in magnetic fields, making radio data a valuable tool for estimating magnetic field strengths. Determining these magnetic fields is essential for understanding particle acceleration processes and the evolution of shock waves in astrophysical environments.

A widely used approach for estimating magnetic fields is the equipartition method, which assumes that the energy stored in relativistic particles is comparable to the energy stored in the magnetic field [1]. Classical equipartition calculations have been successfully applied to many radio sources; however, they become less reliable when the particle energy spectrum is characterized by an energy index smaller than two. In such cases, modified formulations are required to obtain physically meaningful magnetic-field estimates.

Recent studies have developed adapted equipartition calculations that account for particle spectra with $\Gamma < 2$ and are particularly suitable for supernova remnants [2, 3]. These formulations relate the observed synchrotron emission to the energy densities of relativistic particles and magnetic fields, allowing both amplified and ambient magnetic fields to be estimated from observable quantities.

The objective of this project is to apply the adapted equipartition method to the Large Magellanic Cloud supernova remnant J0450–7050. Using the observational parameters provided in the project data file, the amplified magnetic field and the ambient magnetic field are estimated under both electron and proton equipartition assumptions. The required energy integrals are evaluated numerically, and the resulting nonlinear equations are solved using Python.

2 Theoretical Background

2.1 Synchrotron Emission and Equipartition

The radio emission from supernova remnants is generally interpreted as synchrotron radiation produced by relativistic electrons moving in magnetic fields. The observed synchrotron flux density depends on both the electron energy distribution and the magnetic field strength. For a power-law electron spectrum,

$$N(E) = K_e E^{-\Gamma}, \quad (1)$$

where K_e is the normalization constant and Γ is the particle energy index.

For a synchrotron-emitting source, the flux density can be expressed as [3]

$$S_\nu = \frac{4\pi}{3} f \theta^3 d c_5 K_e (B \sin \Theta)^{(\Gamma+1)/2} \left(\frac{\nu}{2c_1} \right)^{(1-\Gamma)/2}, \quad (2)$$

where f is the volume filling factor, θ is the angular radius, d is the distance to the source, and B is the magnetic field strength. The constants c_1 and c_5 are synchrotron coefficients defined by Pacholczyk [1].

Using the relation

$$\alpha = \frac{\Gamma - 1}{2}, \quad (3)$$

the flux-density expression can be rewritten in the form

$$K_e B^{\alpha+1} = J, \quad (4)$$

where J depends only on observable quantities and can therefore be calculated directly from the observational data [3].

The equipartition approach assumes that the energy density of relativistic particles is comparable to the magnetic-field energy density. This assumption provides a connection between K_e and B , allowing the magnetic field strength to be estimated from radio observations.

2.2 Electron and Proton Equipartition

The energy density of relativistic particles can be obtained by integrating the particle momentum distribution. For a power-law distribution of cosmic-ray electrons, the energy density is given by [2, 3]

$$\epsilon_{\text{CR}} = K_e (m_e c^2)^{2-\Gamma} I_1, \quad (5)$$

where

$$I_1 = \int_{x_{\text{inj}}}^{x_{\infty}} x^{-\Gamma} \left(\sqrt{x^2 + 1} - 1 \right) dx \quad (6)$$

is a dimensionless integral, with

$$x = \frac{p}{m_e c}. \quad (7)$$

The lower integration limit is determined by the injection momentum,

$$p_{\text{inj}} = \xi \sqrt{2m_e k T_e}, \quad (8)$$

where ξ is the injection parameter. The upper limit is determined by the maximum electron energy, which is limited by synchrotron losses [4].

Assuming equipartition between the energy density of cosmic-ray electrons and the magnetic-

field energy density,

$$\epsilon_{\text{CR}} = \epsilon_B = \frac{B^2}{8\pi}, \quad (9)$$

the normalization constant can be written as

$$K_e = \frac{(m_e c^2)^{\Gamma-2}}{8\pi I_1} B^2. \quad (10)$$

Substituting this expression into the synchrotron emissivity relation leads to the electron equipartition magnetic-field equation

$$B^{\alpha+3} = 8\pi (m_e c^2)^{2-\Gamma} I_1 J. \quad (11)$$

An alternative approach assumes equipartition between the magnetic field and relativistic protons. In this case,

$$K_p = \frac{(m_p c^2)^{\Gamma-2}}{8\pi I_2} B^2, \quad (12)$$

where I_2 is defined in the same manner as I_1 , but using

$$x = \frac{p}{m_p c}. \quad (13)$$

The corresponding proton equipartition magnetic field is obtained from

$$B^{\alpha+3} = 8\pi (m_p c^2)^{2-\Gamma} \frac{I_2 J}{K_{ep}}, \quad (14)$$

where K_{ep} is the electron-to-proton ratio [5]. In this project, both electron and proton equipartition assumptions are considered and the resulting magnetic-field estimates are compared.

3 Numerical Method

The calculations were performed using Python. The observational parameters for the supernova remnant J0450–7050 were taken directly from the data file provided with the project. The adopted parameters are listed in Table 1.

All quantities were converted to the cgs system before the calculations were performed. The particle energy index was obtained from the observed spectral index using

$$\Gamma = 2\alpha + 1. \quad (15)$$

Table 1: Input parameters used in the calculations.

Parameter	Value
Spectral index, α	0.26
Observing frequency, ν	1 GHz
Flux density, S_ν	0.66 Jy
Distance, d	50 kpc
Angular radius, θ	3.8 arcmin
Filling factor, f	0.25
Shock velocity, v_s	170 km s ⁻¹
Sound speed, c_s	10 km s ⁻¹
Injection parameter, ξ	4
Electron-to-proton temperature ratio, β	1

The shock Mach number and compression ratio were then calculated from

$$M_1 = \frac{v_s}{c_s}, \quad (16)$$

and

$$\sigma = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2}. \quad (17)$$

Using these quantities, the parameter J was evaluated from the synchrotron emissivity relation. The dimensionless integrals I_1 and I_2 appearing in the electron and proton equipartition formulations were computed numerically using the adaptive quadrature routine available in the SciPy library.

The electron and proton equipartition equations are nonlinear because the integration limits depend on the magnetic field strength. Consequently, the magnetic field cannot be obtained analytically. Instead, the corresponding equipartition equations were solved numerically using the Brent root-finding algorithm implemented through the `root_scalar` routine in SciPy.

After determining the amplified magnetic field B , the ambient magnetic field was obtained from

$$B_0 = \kappa B, \quad (18)$$

where κ depends on the shock compression ratio. The final calculations were carried out separately for the electron and proton equipartition assumptions.

4 Results and Discussion

Using the observational parameters listed in Table 1, the adapted equipartition calculations were performed under both electron and proton equipartition assumptions. The resulting amplified and ambient magnetic fields are summarized in Table 2.

Table 2: Calculated magnetic fields for J0450–7050.

Method	Amplified Field (μG)	Ambient Field (μG)
Electron equipartition	24.12	7.35
Proton equipartition	77.38	23.57

For comparison, the reference values provided in the project description are listed in Table 3.

Table 3: Comparison with the reference values provided in the project description.

Quantity	Calculated (μG)	Reference (μG)
B_e	24.12	20.0
$B_{0,e}$	7.35	6.1
B_p	77.38	68.9
$B_{0,p}$	23.57	21.0

The calculated magnetic fields are in reasonable agreement with the reference values. The electron equipartition calculation produces a lower magnetic-field estimate than the proton equipartition calculation, reflecting the different assumptions regarding the dominant cosmic-ray population. In both cases, the amplified magnetic field is significantly larger than the corresponding ambient magnetic field, indicating substantial magnetic-field amplification at the shock.

The proton equipartition method yields amplified and ambient magnetic fields approximately three times larger than those obtained from electron equipartition. This behaviour is expected because relativistic protons carry a much larger fraction of the total cosmic-ray energy budget. Consequently, a stronger magnetic field is required to satisfy the equipartition condition.

The differences between the calculated and reference values are relatively small and may arise from numerical approximations, integration limits, or differences in the implementation of synchrotron constants and pitch-angle averaging. The electron equipartition calculation produced magnetic fields that are approximately 20% larger than the reference values, while the proton equipartition calculation differed by about 12%. Despite these differences, the calculated fields remain in good agreement with the expected values and reproduce the overall behaviour of the adapted equipartition model. Therefore, the overall agreement confirms that the numerical procedure successfully reproduces the adapted equipartition calculation for J0450–7050.

5 Conclusion

In this project, the adapted equipartition method was applied to estimate the amplified and ambient magnetic fields of the Large Magellanic Cloud supernova remnant J0450–7050. Using the observational parameters provided in the data file, the synchrotron emissivity relation and equipartition assumptions were used to derive the magnetic-field estimates under both electron and proton equipartition scenarios.

The numerical calculations yielded amplified and ambient magnetic fields of 24.12 μG and 7.35 μG , respectively, for electron equipartition, while the proton equipartition calculation

produced $77.38 \mu\text{G}$ and $23.57 \mu\text{G}$. The obtained values are in reasonable agreement with the reference values provided in the project description.

Overall, the results demonstrate that adapted equipartition calculations provide a practical method for estimating magnetic fields in supernova remnants with particle energy indices smaller than two. The project also illustrates the application of numerical integration and root-finding techniques to solve nonlinear astrophysical problems.

A Numerical Implementation

A.1 Physical Constants and Input Parameters

```
1 import numpy as np
2 from scipy.special import gamma as GammaFunc
3 from scipy.integrate import quad
4 from scipy.optimize import root_scalar
5
6 c = 2.99792458e10
7 e = 4.803204712e-10
8 m_e = 9.10938356e-28
9 m_p = 1.6726219e-24
10 k_B = 1.380649e-16
11
12 alpha = 0.26
13
14 S_nu_Jy = 0.66
15
16 nu = 1.0e9
17
18 distance_kpc = 50.0
19
20 theta_arcmin = 3.8
21
22 f = 0.25
23
24 vs_kms = 170.0
25
26 cs_kms = 10.0
27
28 xi = 4.0
29
30 beta = 1.0
```

Listing 1: Physical constants and observational parameters used in the calculation.

A.2 Unit Conversions

```
1 # Parsec
2 p_c = 3.085677581e18
3
4 # Flux density
5 S_nu = S_nu_Jy * 1e-23
6
7 # Distance
8 d = distance_kpc * 1e3 * p_c
9
10 # Angular radius
11 theta = theta_arcmin * np.pi / (180.0 * 60.0)
12
13 # Velocities
14 vs = vs_kms * 1e5
15
16 cs = cs_kms * 1e5
```

Listing 2: Conversion of observational parameters to cgs units.

A.3 Shock Parameters and Particle Temperatures

```
1 Gamma = 2*alpha + 1
2
3 M1 = vs/cs
4
5 gamma = 5.0/3.0
6
7 sigma = (
8     (gamma + 1)*M1**2
9     /
10    (2 + (gamma - 1)*M1**2)
11 )
12
13 kappa = 1.0 / np.sqrt(
14     (1 + 2*sigma**2)/3
15 )
16
17 Tp = (
18     ((gamma + 1 + 2*gamma*(M1**2 - 1))
19     *
20     (2 + (gamma - 1)*M1**2))
21     /
22     (gamma*(gamma + 1)**2*M1**2)
23 ) * (m_p*cs**2/k_B)
24
25 Te = beta*Tp
```

Listing 3: Calculation of shock parameters and particle temperatures.

A.4 Calculation of the Synchrotron Parameter

```
1
2
3 pitch_avg = (
4     np.sqrt(np.pi)/2
5     *
6     GammaFunc((Gamma + 5)/4)
7     /
8     GammaFunc((Gamma + 7)/4)
9 )
10
11 c1 = (
12     3*e
13     /
14     (4*np.pi*m_e**3*c**5)
15 )
16
17 c5 = (
18     np.sqrt(3)
19     * e**3
20     /
21     (16*pi*m_e*c**2)
22 )
23
24 c5 *= (
25     GammaFunc((3*Gamma - 1)/12)
26     *
27     GammaFunc((3*Gamma + 7)/12)
28 )
29
30 c5 *= (
31     2/(Gamma + 1)
32 )
33
34 J = (
35     S_nu
36     /
37     (
38         (4*np.pi/3)
39         * f
40         * theta**3
41         * d
42         * c5
```

```

43         * pitch_avg
44         * (nu/(2*c1))**((1-Gamma)/2)
45     )
46 )
47 eta = (
48     4/np.sqrt(np.pi)
49     *
50     (1/(Gamma - 1))
51     *
52     xi**3
53     *
54     np.exp(-xi**2)
55 )

```

Listing 4: Calculation of the synchrotron parameter J.

A.5 Electron Equipartition Calculation

```

1  def I1_of_B(B):
2
3      pinj = xi*np.sqrt(2*m_e*k_B*Te)
4
5      Einf = (
6          ((sigma - 1)/(3*sigma))
7          *
8          (m_e**2*c**3*vs)
9          /
10         (
11             (1 + np.sqrt(kappa))
12             *
13             np.sqrt((2/27)*e**3*B)
14         )
15     )
16
17     xinj = pinj/(m_e*c)
18
19     xmax = Einf/(m_e*c**2)
20
21     integrand = lambda x: (
22         x**(-Gamma)
23         *
24         (np.sqrt(x**2 + 1) - 1)
25     )
26
27     value, _ = quad(
28         integrand,
29         xinj,
30         xmax

```

```

31     )
32
33     return value
34
35 def electron_equation(B):
36
37     I1 = I1_of_B(B)
38
39     rhs = (
40         8*np.pi
41         *
42         (m_e*c**2)**(2-Gamma)
43         *
44         I1
45         *
46         J
47     )
48
49     return B**(alpha + 3) - rhs
50
51 sol = root_scalar(
52     electron_equation,
53     bracket=[1e-7, 1e-3]
54 )
55
56 B_e = sol.root
57
58 B0_e = kappa*B_e
59
60 I1 = I1_of_B(B_e)
61
62 pinj = xi*np.sqrt(2*m_e*k_B*Te)

```

Listing 5: Evaluation of the electron equipartition magnetic field.

A.6 Proton Equipartition Calculation

```

1 R = d*theta/2
2
3 def I2_of_B(B):
4
5     B0 = kappa*B
6
7     pinj = xi*np.sqrt(
8         2*m_p*k_B*Tp
9     )
10
11     Einf = (

```

```

12         3*vs*e*B0*R
13         /
14         (8*c)
15     )
16
17     xinj = pinj/(m_p*c)
18
19     xmax = Einf/(m_p*c**2)
20
21     integrand = lambda x: (
22         x**(-Gamma)
23         *
24         (np.sqrt(x**2 + 1) - 1)
25     )
26
27     value, _ = quad(
28         integrand,
29         xinj,
30         xmax
31     )
32
33     return value
34
35 def Kep():
36
37     return (
38         (m_e*Te)
39         /
40         (m_p*Tp)
41     )**alpha
42
43
44 def proton_equation(B):
45
46     I2 = I2_of_B(B)
47
48     rhs = (
49         8*np.pi
50         *
51         (m_p*c**2)**(2-Gamma)
52         *
53         I2
54         *
55         J
56         /
57         Kep()
58     )
59

```

```

60     return B**(alpha + 3) - rhs
61
62 sol_p = root_scalar(
63     proton_equation,
64     bracket=[1e-6, 1e-3]
65 )
66
67 B_p = sol_p.root
68
69 B0_p = kappa*B_p

```

Listing 6: Evaluation of the proton equipartition magnetic field.

A.7 Final Results

```

1 print("B_e  =", B_e*1e6, "microG")
2 print("B0_e =", B0_e*1e6, "microG")
3
4 print("B_p  =", B_p*1e6, "microG")
5 print("B0_p =", B0_p*1e6, "microG")

```

Listing 7: Final magnetic-field estimates.

References

- [1] Pacholczyk, A. G., *Radio Astrophysics: Nonthermal Processes in Galactic and Extragalactic Sources*, W. H. Freeman and Company, San Francisco, 1970.
- [2] Arbutina, B., Urošević, D., Andjelić, M. M., Pavlović, M. Z., and Vukotić, B., “A New Equipartition Formula for Supernova Remnants,” *The Astrophysical Journal*, 746, 79 (2012).
- [3] Arbutina, B., “Adapted Equipartition Calculation for Supernova Remnants,” *Publications of the Astronomical Observatory of Belgrade*, 96, 1–10 (2017).
- [4] Zirakashvili, V. N. and Aharonian, F. A., “Analytical Solutions for Energy Spectra of Electrons Accelerated by Nonrelativistic Shock Waves in Shell-Type Supernova Remnants,” *Astronomy & Astrophysics*, 465, 695–702 (2007).
- [5] Arbutina, B. and Zeković, V., “Equipartition Calculations and Magnetic Field Estimates in Supernova Remnants,” *Astrophysics and Space Science*, 2021.